4 **(a)** Define the resistance of a circuit component.

.....

..... [1]

(b) With reference to energy transfers, distinguish between electromotive force (e.m.f.) and potential difference (p.d).

.....

.....

..... [1]

(c) Fig. 4.1 shows a circuit containing cell of e.m.f. E with internal resistance r, a fixed resistor with resistance R_1 and a variable resistor with resistance R_2.

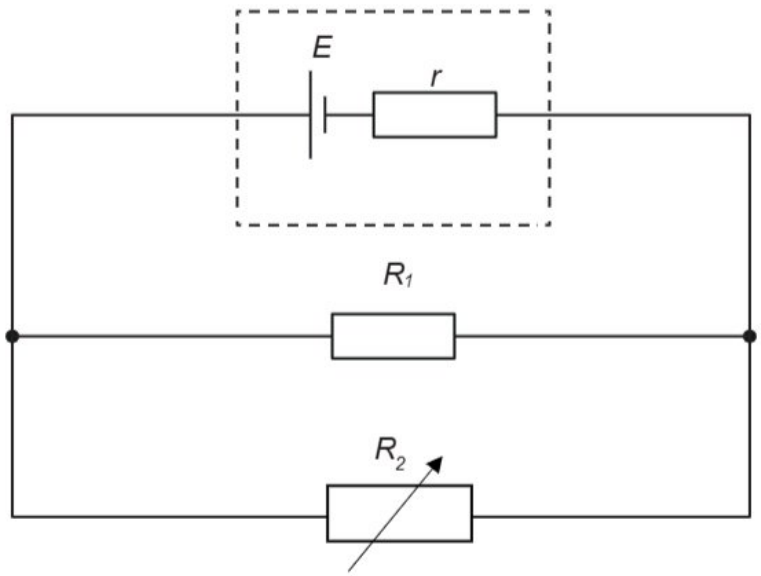


Fig. 4.1

With reference to the current in the cell, explain what happens to the current through R_1 when R_2 increases.

.....

.....

.....

.....

.....

..... [3]

(d) The e.m.f. E of the cell in Fig. 4.1 is 1.40 V. The internal resistance r of the cell is 0.13 Ω.

R_1 has a resistance of 5.50 Ω.

At a particular resistance R_2, the current through R_1 is measured to be 0.220 A.

Determine

(i) the current through the cell

current = ... A [2]

(ii) the resistance R_2.

R_2 = ... Ω [2]

[Total: 9]